Sparse Feature Clustering Network for Unsupervised SAR Image Change Detection

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***Abstract*— In this article, we propose a sparse feature clustering network (SFCNet) for change detection in synthetic aperture radar (SAR) images. One of the principal problems in dealing with SAR images is to reduce the impact of speckle noise. Therefore, based on a neural network framework for change detection, we introduce the multiobjective sparse feature learning (MO-SFL) model where the sparsity of representation is adaptively learned in order to increase the robustness to different levels of noise. For learning the semantic information of changed and unchanged pixels, the network is fine-tuned by the correctly labeled samples selected from coarse results. The selection criterion influences the change detection result a lot. Therefore, we construct a novel cross-entropy clustering loss (CEC) by introducing a clustering regularization term to learn the discriminative representations. Experiments on simulate and real SAR images demonstrate the superiority of the proposed method over compared methods.**

***Index Terms*— Deep neural network (DNN), image change detection, representation learning, synthetic aperture radar (SAR).**

I. Introduction

IMAGE change detection is a process in which changed regions between two images of the same scene but taken at different times are identified. It has attracted wide attention in the applications of remote sensing [1]–[16], medical diagnosis [17], and video surveillance [18], [19]. With the development of remote sensing technology, large parts of the Earth's surface have been observed by remote sensing sensors. Change detection in remote sensing images is of great importance, especially for disaster evaluation. Among remote sensing images, synthetic aperture radar (SAR) images are independent of solar illumination and atmospheric condition. In this article, we focus on single-band SAR images that are one of the useful and indispensable sources of information in change detection [9]–[12], [20]. However, with the influence of speckle noise, change detection in SAR images is more challenging than in optical images.

After coregistration, the general procedure of unsupervised change detection can be divided into two main steps: the first step is to produce the difference image (DI) between the multitemporal images and the second step is the analysis of the DI [21], [22]. The second step is similar to the process of image segmentation. Thus, two common approaches, the thresholding and clustering approaches, are widely used. The thresholding approaches, such as Otsu [23] and the Kittler and Illingworth [24], usually establish appropriate distribution models to search for the best threshold to divide the DI into two classes. However, it is difficult to estimate the distribution of these two classes. Clustering methods, such as the fuzzy c-means (FCM) algorithm [25] and the reformulated fuzzy local-information c-means (RFLICM) algorithm [26], do not need to establish a distribution model and are more flexible. Besides these methods, many other approaches, such as the Markov random field (MRF) based [27] and graph cuts based [28], have been proposed for change detection in SAR images.

In SAR images, speckle noise is a common and resistant noise, which greatly influences the performance of image interpretation approaches [29]–[33]. The above methods analyze the properties of SAR images manually to reduce the influence of speckle noise. Moreover, because the satellite is not consistent between two acquisition times and the posture of the satellite influences the quality of the SAR image a lot, different SAR images may have different numbers of looks. This means that the SAR images may have inconsistent noise levels. With the development of machine learning, recently many learning-based methods have been proposed for unsupervised SAR image change detection. Gong *et al.* did the seminal work in [34] where an end-to-end deep neural network (DNN)

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was trained with selected labeled samples. Based on this framework, many unsupervised methods have been proposed. A ternary change detection method was proposed in [20], which focuses on two types of changes. A convolutional neural network (CNN)-based method was proposed in [12], where the spatial characteristics of the raw image can be extracted and captured by automatic learning, and the results with stronger robustness can be obtained. In addition to neural networks, other learning methods, such as self-paced learning [35], were also designed for change detection based on the learning framework in [34].

Recent works have paid more attention to the sparse representations for image denoising and despeckling [36]–[38], which have been demonstrated to be effective. As an important type of model in the development of deep learning, sparse feature learning models [39], [40] were proposed to pretrain robust DNNs. The common way is to add sparse constraints to the base models, e.g., autoencoder (AE) and restricted Boltzmann machine (RBM). For instance, based on AE, an energy model was proposed in [39] by transforming a code vector into a sparse code vector. Based on RBM, a sparse response RBM (SR-RBM) was proposed in [40] by adding a sparse response constraint to the Kullback–Liebler (KL) divergence in the RBM. However, to learn reasonable sparse representations, the importance of the sparse constraint term is controlled by a user-defined parameter. With different input data and applications, the optimal parameter is also different. This limits the application of these feature extractors to heterogeneous data. Later, Gong *et al.* proposed a multiobjective sparse feature learning (MO-SFL) approach in [41] where minimizing the sparsity and reconstruction error are taken as two objectives and optimized simultaneously. It solves a conflicting problem by converting it into a multiobjective optimization problem [42]–[46]. By analyzing the Pareto solutions generated via the multiobjective optimization method, an adaptive solution can be obtained. This approach is suitable to deal with the inconsistent noise of SAR images. Since the noise is the redundant information, with different noise levels, the optimal sparsity of the features is also different. Therefore, we use MO-SFL to pretrain the proposed network in order to increase the robustness of the network to SAR images with different numbers of looks.

Another problem in the change detection framework in [34] is the influence of training samples. To train the network without manually labeled samples, some correctly classified samples have to be selected from the results of an existing method to learn the semantics of changed and unchanged regions [12], [20], [34], [35]. However, with the absence of prior information, the correctly classified samples are approximated based on the neighborhood consistency of the result. Strict criteria can ensure the correctness of the samples while reducing the number of samples greatly. This will influence the fine-tuning of the network, which influences the robustness to corruptions, while relaxed criteria introduce more wrongly classified samples, which mislead the network. Focusing on this problem, we introduce a clustering-based regularization term to the objective function. The regularization term is designed based on the c-means clustering, which is widely used in SAR image change detection as introduced above. Then, the fine-tuning of the network depends not only on the selected correctly labeled samples but also on the unselected samples. This regularization term can avoid the uncertainty of samples' selection and encourage the neural network to learn more interpretable and compact representations. As a consequence, by combining MO-SFL and the clustering regularization-based learning, we propose the sparse feature clustering network (SFCNet) for unsupervised change detection in SAR images. The major contributions of this article are the following three main aspects.

1) An SFCNet is proposed for change detection in SAR images.
2) The MO-SFL theory is introduced into the construct learning-based framework to automatically increase the change detection robustness to speckle noise.
3) A new cross-entropy clustering loss (CEC) is proposed by adding the clustering regularization term on the cross-entropy fine-tuning loss to further improve the discriminative performance for SAR image change detection.
4) Experimental results on simulate and both small- and large-scale real SAR images demonstrate the superiority of the SFCNet by comparing it with state-of-the-art methods.

The rest of this article is structured as follows. In Section II, we review the neural network framework for change detection in SAR images, sparse feature learning, and the motivation for proposing the SFCNet. In Section III, the SFCNet is described in detail. Experimental results are reported and analyzed in Section IV. Finally, Section V draws the conclusion and future work of this article.

II. Related Work and Motivation

A. Learning-Based Change Detection in SAR Images

The development of representation learning [47] has revealed the learning capability of DNNs. Then, based on the learning paradigm of the deep belief neural network (DBN) [48], an SAR image change detection by training a neural network was first proposed in [49]. Later, the same group developed a DNN framework for end-to-end SAR image change detection in [34]. The process of this change detection framework can be divided into three steps. First, for unsupervised change detection, there are no manually labeled samples to train the DNN. The labeled samples can be obtained by existing methods. However, the result of them cannot be guaranteed to be completely correct. Therefore, reliable samples with high confidence in correct classification are first chosen. In [34], a joint FCM was used to classify the two images by the similarity of the gray levels, and the classification results are compared to obtain the coarse change detection result. The reliable labeled samples are selected for fine-tuning the network. Secondly, the network parameters are initialized by feature learning models, and the RBM is used to pretrain the DNN. The input of the network is the neighborhood of each pixel in the two corresponding images. After pretraining, the representation of the images can be

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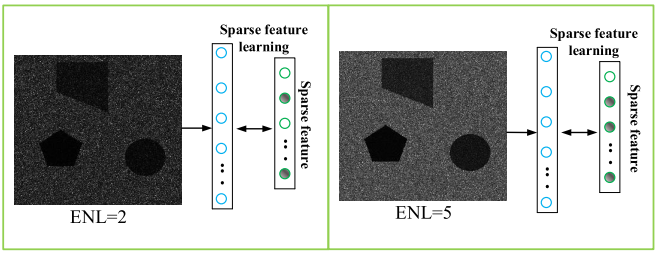


Fig. 1. Different degrees of sparsity for different noise levels.

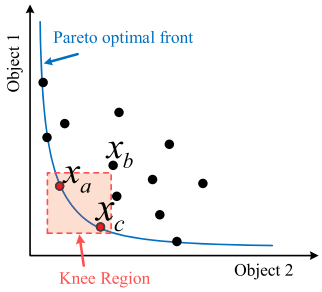


Fig. 2. PF and solutions.

extracted. Finally, the network is fine-tuned in order to learn the semantics of changed and unchanged regions. The labeled samples selected in the first step are used to fine-tune the network via the cross-entropy classification loss. Finally, with the input of two original images, the network is used to classify each pixel into changed or unchanged ones.

However, the framework follows the basic learning paradigm of DBN, which does not especially consider the noise in SAR images. To overcome this difficulty, in this article, we improve the framework by introducing the sparse feature learning theory. RBM and AE have demonstrated promising results in representation learning [47], [50], [51]. Without additional constraints, RBM and AE may produce redundant features and unstructured weight patterns. Therefore, RBM and AE may not well capture the structure of the data when corrupted severely by noise. Among the constraints, sparsity is popular in many applications [40]. Sparsity is able to facilitate the model to learn succinct features and structured weight patterns. In order to deal with varied speckle noise for change detection in SAR images, we introduce an MO-SFL model in the DNN framework to learn the feature representation that is robust to the variance of the noise. As introduced above, the SAR images are usually corrupted by different noise levels with different acquisition times. Then, the degree of sparsity for different datasets is also different. As shown in Fig. 1, two emulational SAR images with different equivalent numbers of looks (ENL) are exhibited. With lower ENL, the image is corrupted more by noise, and the features should be more sparse to abstract useful information. Most traditional sparse feature learning models use a user-defined parameter to control the sparsity. Therefore, it has to be adjusted for each dataset in order to achieve optimal performance. Therefore, the MO-SFL is introduced in our network to adaptively obtain reasonable sparsity. This is suitable to deal with different noise levels of SAR images.

B. Multiobjective Sparse Feature Learning

From sparse representation theory, images can be taken as a combination of noise and contents in images. Since noise is random, which cannot be encoded as sparse codes, the component of noise can be removed in the sparse representation [52]. As a consequence, sparse representation is an important method for image denoising [53], [54]. In neural networks, sparse feature learning is inspired by the simple cells in the neural cortex, which aims to generate a sparse code with primal information to represent input data while removing redundant information [39], [40], [55], [56]. With the denoising capability of sparse feature learning, it is widely applied to dealing with SAR images [36]–[38].

In [41], an MO-SFL model was proposed for pretraining DNNs. It models the sparse feature learning problem into a multiobjective model to adaptively learn the sparsity. The objective function of a multiobjective optimization is defined as a vectorial model

$$\min_{x} \; F^M(x) = (f_1(x), f_2(x), \ldots, f_m(x), \ldots, f_M(x))^T \tag{1}$$

where $x \in \Phi$ and Φ is the feasible region. F includes M objective functions, and x is the solution in the decision space. With two solutions x_a and x_b, x_a is said to dominate x_b if

$$\begin{aligned} &\forall\, m = 1, 2, \ldots, M, \quad f_m(x_a) \le f_m(x_b) \\ &\exists\, m = 1, 2, \ldots, M, \quad f_m(x_a) < f_m(x_b) \end{aligned} \tag{2}$$

which can be written as $x_a \prec x_b$. If neither dominates the other between solution x_a and solution x_b, this case is defined as solutions that are nondominated by each other. If there is no solution that can dominate x_a, x_a is one optimal solution on the Pareto optimal front (PF). PF is a boundary between feasible and infeasible regions. Fig. 2 illustrates the PF and nondominated solutions in which x_a and x_c are nondominated with each other, and they both dominate x_b. The goal of the multiobjective optimization algorithm is to output a wide variety of optimal solutions on PF, as shown in Fig. 2. After optimized by a multiobjective evolutionary algorithm (MOEA), a set of solutions on the PF is obtained. We need to select a solution from the set with user preference or the property of the PF. The knee region, as shown in Fig. 2, is an area in which the decrease in an objective will increase the other objectives sharply. Solutions in such areas usually demonstrate better compromise between the conflicting objectives. Therefore, users prefer solutions in the knee regions.

MO-SFL was proposed based on AE. AE is a typical feature learning model with encoder–decoder architecture and can be stacked to pretrain a DNN. With the encoder $f(\cdot)$, decoder $g(\cdot)$, and the corresponding parameter set θ, including connecting weights and biases, the learning of AE is to optimize the following objective:

$$\min_{\theta} \; J_{\mathrm{AE}}(\theta) = \sum_{x \in D} L(x, g(f(x)) \tag{3}$$

where D is the training dataset and the reconstruction error $L(x, g(f(x)))$ is the mean square error between the

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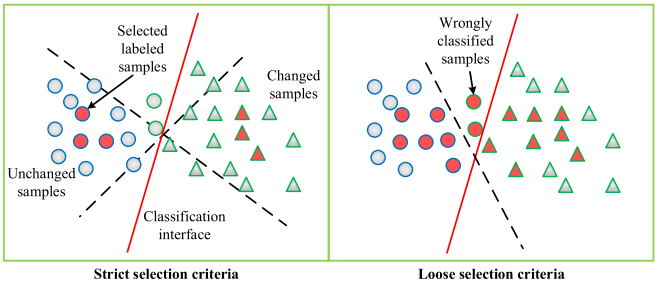


Fig. 3. Illustration of classification interfaces under different conditions.

two vectors $\|x - g(f(x))\|^2$. By considering the sparsity $\sum_{x\in D}\|f(x)\|_0$ as another objective, the MO-SFL is obtained

$$\min_{\theta} \; F_{\text{MO-SFL}}(\theta) = \left(\sum_{x\in D} L(x, g(f(x))), \sum_{x\in D}\|f(x)\|_0\right)^T \tag{4}$$

where the sparsity is defined as the nonzero components in hidden units. To conveniently encode the input data with sparse code, we use the sigmoid function as the activation function of the encoder, which is widely used in sparse feature learning [39], [40], [55], [56]. An efficient learning procedure for the multiobjective model was designed in [41], where the model is optimized by combining the gradient descent algorithm and evolutionary algorithm. A solution in the knee region can be obtained by the learning procedure. With different datasets and applications, the model can learn the network parameter adaptively, which can generate appropriate sparse features without determining the control parameter. Therefore, the method is adaptive and is able to adjust to SAR images with different noise levels.

C. Sample Selection and Objective Function

In the learning-based framework for SAR image change detection, another problem that influences the change detection accuracy is the selected labeled samples. As shown in Fig. 3, some labeled samples with high confidence have to be selected to learn the semantics of changed and unchanged regions. Since the labels are obtained by an unsupervised method, there are many wrongly classified samples, such as the green circles. Here, samples with high confidence mean that the selected samples are likely to be correctly labeled, which is evaluated by the selection criteria. To train a network effectively, it is better to use more correctly labeled samples. Therefore, strict selection criteria can guarantee the correctness of the samples while may be not sufficient to learn the correct classification interface, i.e., the red line. With fewer learning samples, the classification interface learned may be shown as the dashed lines. However, when the selection criteria are loosened, the wrongly classified samples will be selected, which may mislead the classifier as shown by the dashed line. Therefore, it is critical to determine the selection criteria.

In this article, we introduce a new clustering regularization term based on c-means on the fine-tuning loss in order to release the dependence on the labeled samples. The clustering regularization is computed by all the samples, and then, the unselected samples can be used to facilitate the fine-tuning process. The clustering is implemented on the features generated by the network; thus, by adding this regularization term, the network can learn the compact and discriminative representations. The clustering can be implemented on any hidden layer, and here, we use the features generated in the last hidden layer.

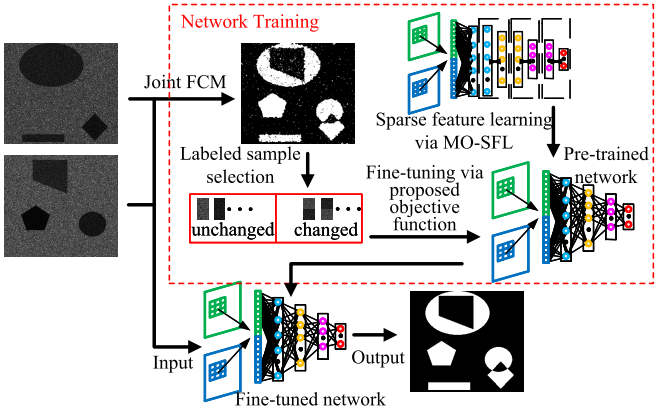


Fig. 4. Change detection process of the proposed SFCNet for SAR images.

III. Sparse Feature Clustering Network

In this section, we introduce and analyze the proposed SFCNet for SAR image change detection in detail. Fig. 4 shows the change detection process of SFCNet, which follows the widely used framework in [34]. The aim of SFCNet is to take change detection as a binary classification problem and train a network to classify changed and unchanged pixels. The training process is composed of three modules: 1) training sample selection to generate the labeled training samples via the coarse change detection results of existing methods; 2) sparse feature learning to learn the sparse feature representations of the neural network and pretrain the network parameters; and 3) fine-tuning via the proposed loss function with selected labeled samples to further tune the network parameters. After training, the labels of all the pixels are assigned by the fine-tuned neural network. Different from the existing learning methods, in this article, we use MO-SFL to adaptively pretrain the network and introduce the clustering regularization for the fine-tuning loss, which makes the network more robust to selected samples.

In this section, the way of training samples' selection is first introduced. Then, the sparse feature learning for pretrain the network is described and analyzed to show its robustness to various noises. Finally, the fine-tuning process by optimizing the proposed objective function is described.

A. Training Samples' Selection

Since the DNN is a data-driven method, labeled samples are necessary to learn the semantics of change and unchange. However, in change detection, the remote sensing images are heterogeneous, and it is difficult to label them manually. Fortunately, with unsupervised feature learning, the DNN

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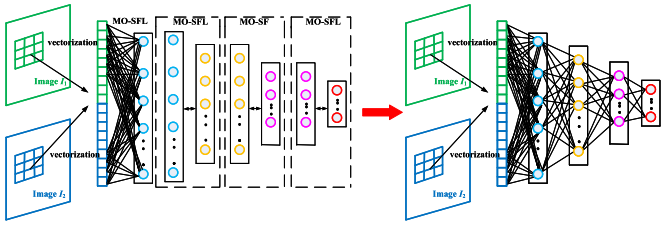


Fig. 5. Structure of the sparse feature learning. With the input local pixels, MO-SFL models are constructed layerwise as shown in the left part. After training, the encoders of the trained MO-SFL models are stacked to form a multilayer neural network.

can achieve promising results with less labeled samples. The labeled samples can be obtained from the result by other methods. Since the labels in the result are not completely correct, we can select some reliable samples with labels of high confidence to train the network. Therefore, the result does not need to be very accurate and could be acquired by some simple and direct methods, such as FCM. Therefore, in this article, we use the joint FCM proposed in [34] to generate a coarse result.

After the change detection result is generated, it is critical to select reliable labeled samples. Here, we select them according to the consistency of local pixels. For each pixel of the result, if the labels of most pixels within the neighborhood region (here we set 90%) are consistent (changed for instance), then the pixel is selected and is assigned as the label of changed. With the severe influence of speckle noise, incorrectly assigned pixels are usually isolated. Therefore, by using the local consistency-based sample selection, reliable labeled samples can be selected.

However, as analyzed in Section II, the sample selection influences the final result a lot. Therefore, in this article, we propose a new objective function for training the neural network. Before being trained by labeled samples, the network is first constructed and pretrained by an unsupervised sparse feature learning model.

B. Sparse Feature Learning

In this article, the loss function of MO-SFL is introduced as the objective function of feature learning, which is able to adapt to various noise levels in heterogeneous SAR images without a user-defined parameter. The feature learning model is stacked layer by layer to construct the DNN architecture.

The input of the network is a vector that consists of neighborhood pixels of the two images, as shown in Fig. 5. We choose an $n \times n$ patch with a center point (i, j) and its neighborhoods $(N_{ij}^{I_1}, N_{ij}^{I_2})$ in two SAR images; the size of neighborhood may influence the performance of the network. N_{ij} is normalized between 0 and 1 and converts to $\overline{N}_{ij}$. The input features $\overline{N}_{ij}$ consist of two patches of features $(\overline{N}_{ij}^{I_1}, \overline{N}_{ij}^{I_2})$

$$\overline{N}_{ij} = \left[\overline{N}_{ij}^{I_1}, \overline{N}_{ij}^{I_2}\right]. \tag{5}$$

Then, the MO-SFL is utilized to pretrain the network layerwise. MO-SFL is based on AE and the architecture consists of an encoder, a decoder, and a sparse term. The objective function of MO-SFL is modeled as an multi-objective optimization problem (MOP), as introduced in Section II. For solving MOPs, MOEAs are used because they can produce the Pareto optimal solutions with a single run by the interaction between the Pareto solutions. However, in a neural network, the scale of parameters that have to be optimized is usually squared over the number of units. For example, with two 9×9 neighborhood patches inputted to the network, the number of visible units is 162, and the number of hidden units is not less than 162. Thus, the connecting weights and bias are more than 162×162. This searching space is large, which results in the difficulty of finding satisfactory solutions on PF [57]. As in [41], we also adopt a multiobjective induced learning procedure to solve this problem by converting MOEA to several local multiobjective optimization (LMO) models. The solution in each LMO is selected from the knee region and is used in the next iteration.

The LMO learning process based on the MO-SFL model includes four steps.

1) Initialize the networks by minimizing the reconstruction error.
2) Use an MOEA method to optimize the biases, and then, choose one Pareto optimal solution for the knee region.
3) Optimize the reconstruction error to update the networks.
4) Return to step 2) until convergence.

In steps 1) and 3), stochastic gradient descent is applied to the optimization of network parameters by minimizing the reconstruction error. After this process, the LMO model is optimized with respect to only the bias vector b because it directly controls the sparsity of hidden units. The MO-SFL model is then divided into several LMO models. The LMO model is easier to be optimized because the scale of b is smaller. Different from the traditional MOEAs, this learning procedure could find one appropriate solution for the knee region instead of a set of Pareto optimal solutions. The new objective function of the LMO model is stated as follows:

$$\min_b \; F_{\mathrm{LMO}}(b) = \left(\sum_{x \in D_s} L\left(f_{\theta^*}(x), f_\theta(x)\right), \sum_{x \in D_s} \| f_\theta(x) \|_1 \right)^T \tag{6}$$

where θ^* is the parameter generated in the last iteration. The set D_s is a stochastic local set with training data. Since the features are encoded by the sigmoid activation function, which can be taken as the probability of the status in each node [40], we use the cross entropy loss, which is widely used in machine learning [58] to measure the distance between $L(f_{\theta^*}(x)$ and $f_\theta(x))$

$$L(f_{\theta^*}(x), f_\theta(x)) = \sum -f_{\theta^*}(x) \log(f_\theta(x)) - \left(1 - f_{\theta^*}(x)\right) \log(1 - f_\theta(x)). \tag{7}$$

The LMO model is optimized by the widely used decomposition-based evolutionary algorithm decomposition-based multi-objective evolutionary algorithm (MOEA/D) [59] in step 2). Evolution algorithms have been proven to be one type of the effective methods for optimizing multiobjective

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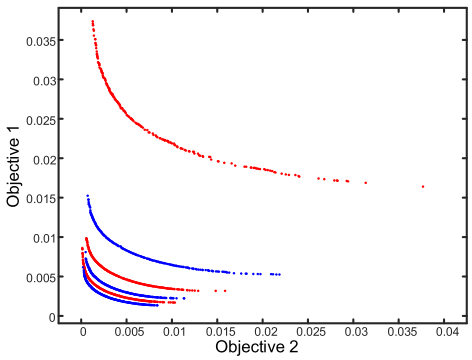


Fig. 6. Pareto solutions in each iteration for solving the multiobjective problem in (6). To distinguish solutions in different iterations, those with adjacent iterations are marked with different colors.

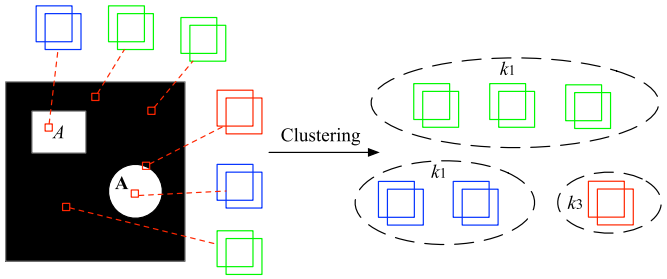


Fig. 7. Illustration of the clustering regularization. Assuming that A is a selected training sample with high confidence and $\mathbf{A}$ is an unselected sample from Section III-A. By adding the clustering regularization constraint, $\mathbf{A}$ can be selected as a training sample furthermore, thereby expanding the high confidence training samples, which is beneficial to the network fine-tuning.

problems [60], [61]. Therefore, MOEA/D is expected to generate the optimal Pareto solutions of the multiobjective problem in (6).

This learning process decomposes the large multiobjective learning in (4) into several small multiobjective learning tasks in (6) by greedy learning. The Pareto solutions in each iteration of step 2) are illustrated in Fig. 6. In each iteration, a set of Pareto solutions can be generated by solving the local multiobjective problem. However, one iteration only can search in a local region. Therefore, in the next iteration, a set of Pareto solutions with smaller objective values are generated. After several iterations, both objectives can be optimized to low values. With the nonconvex property of objectives, the final solution may be not optimal. However, with the multiobjective optimization method, a solution with a better tradeoff between the two objectives can be obtained [41].

After the learning procedure, the network parameters, which are able to generate features with reasonable sparsity, can be obtained for SAR image change detection. Then, another layer is stacked, and higher level features are learned. With the layerwise pretraining, the robust parameters can be initialized, which facilitates the fine-tuning process. After being pretrained, the encoders of them are stacked to form a multilayer neural network, as shown in Fig. 5. This network will be fine-tuned by the proposed objective function.

C. Fine-Tuning via the Clustering Facilitated Loss

In the fine-tuning process, the labeled samples are necessary to learn the semantics of changed and unchanged regions. With the unsupervised change detection, the labeled samples are selected from the result generated by other sample selection methods. Therefore, the labels of the samples may be not completely correct. The selection criteria are important to ensure the correctness of the labels for training and, meanwhile, guarantee the robustness to noise and unexpected corruptions. Then, the change detection result is usually sensitive to the selection criteria. In this article, we introduce a clustering regularization to the fine-tuning loss to alleviate this problem.

With an input sample, the features can be generated at the hidden layers. Let A^q denote the feature vector generated at a hidden layer by sample q; the clustering-based regularization is then formulated by

$$R_A = \frac{1}{2Q \cdot K} \sum_{q=1}^{Q} \sum_{k=1}^{K} T_k^q \left\| A^q - \overline{B}_k \right\|_2^2 \tag{8}$$

where Q and K denote the number of samples and cluster centers, respectively. $T_k^q = 0, 1$ with 1 denotes that the qth sample belongs to the kth cluster. $\overline{B}_k$ represents the clustering center of the kth cluster. This regularization term is defined based on the c-means clustering algorithm with the purpose of improving the efficiency to use samples. C-means clustering is one of the popular methods to deal with change detection in SAR images [26], [27] due to its flexibility in using different information, such as local correlations [27]. Fig. 7 illustrates the clustering of the samples. The cluster is implemented on the feature of each sample. Even though c-means assumes the Gaussian distribution of input data, with the neural network, the features can be learned and transformed to adapt to the data and objective. The labeled samples provide the seminal clustering solutions. Then, with the clustering-based regularization, unlabeled samples can be used to facilitate the network to learn discriminative and compact features. This can further improve the change detection performance.

Note that R_A is the clustering regularization added to the fine-tuning loss function L_A, which is the cross-entropy error as in [34]. Then, the objective function $\overline{L}_A$ for fine-tuning in SFCNet can be obtained as follows:

$$\overline{L}_A = L_A + \lambda_A R_A \tag{9}$$

where λ_A adjusts the importance of the clustering regularization. The regularization is implemented on the features in hidden layers, and any hidden layer can be used. In this article, we use the last hidden layer that contains abstract and discriminative information. Then, the new objective is optimized by the widely used backpropagation algorithm. Then, we derive the gradient of the regularization term R_A with respect to the feature vector A^i

$$\frac{\partial R_A}{\partial A^i} = \frac{1}{Q \cdot K} \sum_{k=1}^{K} T_k^i \left(A^i - \overline{B}_k \right) - \frac{1}{Q \cdot K} \sum_{q=1}^{Q} \sum_{k=1}^{K} T_k^q (A^q - \overline{B}_k) \frac{1}{Q_k} T_k^i \tag{10}$$

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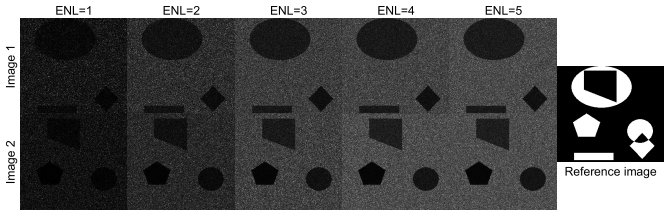


Fig. 8. Simulate dataset where images are generated with different NL values.

where Q_k is the number of the samples that belong to the kth cluster. Then, the gradient is backpropagated through the network, and the gradient of the network is computed.

Note that the clustering centers $\overline{B}_k$ and classes of each sample T_k^q have to be updated along with the network parameters. The network parameters and the clustering parameters are updated alternately as many optimization problems. $\overline{B}_k$ and T_k^q are updated according to the c-means clustering algorithm as follows:

$$T_k^q = \begin{cases} 1, & \text{if } \underset{k}{\operatorname{argmin}} \left\| A^q - \overline{B}_k \right\|^2 = k \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

$$\overline{B}_k = \frac{1}{Q_k} \sum_{q=1}^{Q} T_k^q A^q. \tag{12}$$

After T_k^q and $\overline{B}_k$ have been updated, the network parameters are then updated according to the gradient produced by the backpropagated error. Then, the features are generated, and the clustering parameters are updated again. The two steps are iterated until the stop criterion has been satisfied. Finally, the original images are input into the network, and the change detection result is obtained at the output layer.

IV. Experimental Study

A. Datasets and Evaluation Criteria

To fully evaluate the proposed method, three types of SAR images are used in the experiments. The first type of dataset is the simulate dataset shown in Fig. 8, which is generated by setting different ENLs. Five pairs of images with ENL = 1–5 are generated, which can represent the level of noise. The last one is the reference image, which indicates the changed region with white pixels. This dataset is used to demonstrate the denoising capability of MO-SFL by comparing it with several sparse learning methods.

Then, we choose five small scenes from two large datasets. The first dataset is the Yellow River dataset taken by Radarsat-2 at the region of Yellow River Estuary in China and consists of two 7666×7692 images in June 2008 and June 2009. The Yellow River dataset is widely used in many change detection methods [26], [34]. Two scenes of inland water and farmland are selected, as shown in Fig. 9. The size of images of inland water is 444×291, and that of the farmland is 291×306. The reference images follow the images of the two scenes, respectively, which are generated via integrating prior information (investigation) with photo interpretation based on the input images [26], [34]. The second dataset is taken from

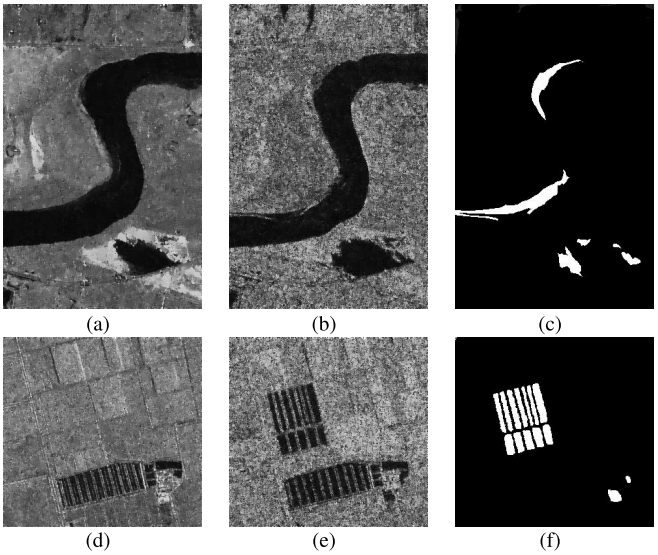


Fig. 9. Two scenes of the Yellow River dataset. Image of inland river acquired in (a) June 2008 and (b) June 2009. (c) Reference image of the scene of the inland river. Image of farmland acquired in (d) June 2008 and (e) June 2009. (f) Reference image of the scene of farmland.

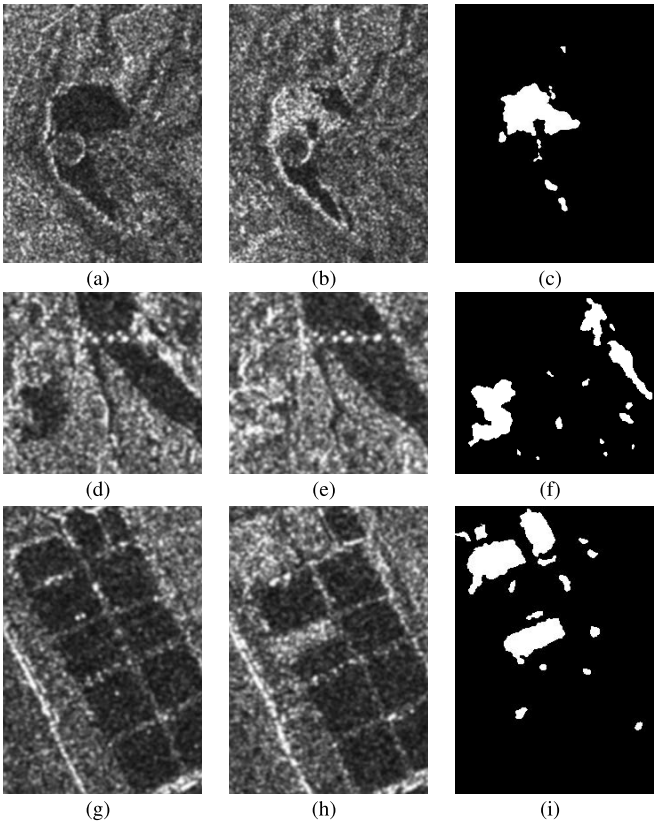


Fig. 10. Three scenes of the Xi'an dataset. Image of river A acquired in (a) March 2010 and (b) March 2013. (c) Reference image of the scene of river A. Image of river B acquired in (d) March 2010 and (e) March 2013. (f) Reference image of the scene of river B. Image of farmland acquired in (g) March 2010 and (h) March 2013. (i) Reference image of the scene of farmland.

an area of Shaanxi, China, the city of Xi'an. The former and latter SAR images are provided by TerraSAR in March 2010 and March 2013, respectively, with a resolution of 3.5 m. We select three scenes of city rivers and farmland, as shown

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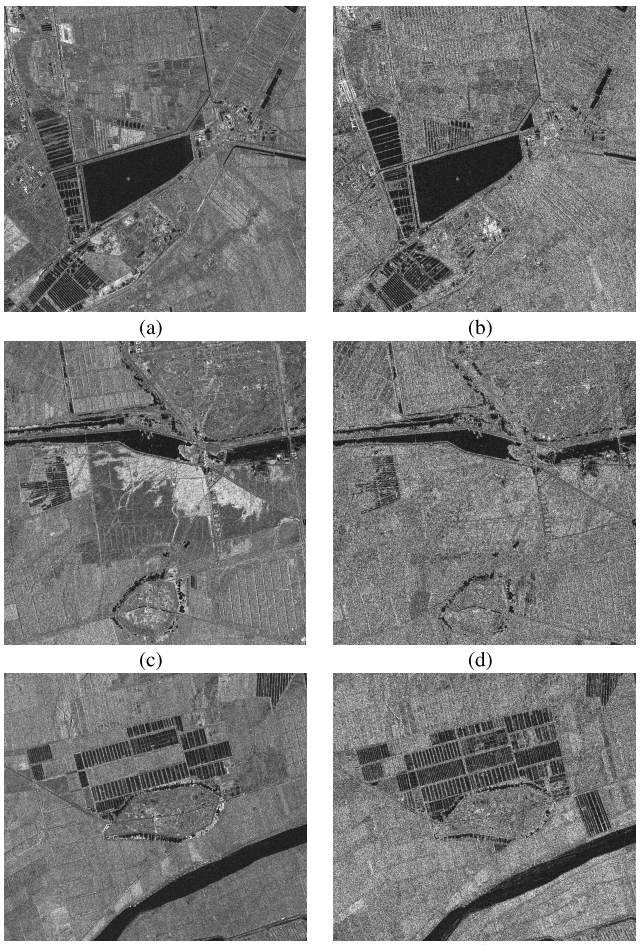


Fig. 11. Three large-scale scenes of the Yellow River dataset. Image of (a) river A acquired in June 2008, (b) scene A acquired in June 2009, (c) scene B acquired in June 2008, (d) scene B acquired in June 2009, (e) scene C acquired in June 2008, and (f) scene C acquired in June 2009.

in Fig. 10. The size of images of river A is 301×391, that of river B is 226×206, and that of the farmland is 206×226. Here, we focus on small patches with known changed regions in real scenes to evaluate the proposed method accurately.

Finally, three large-scale scenes are selected from the Yellow River dataset, as shown in Fig. 11. The image sizes of them are 1207×1221, 1205×1209, and 1239×1095, respectively. With the large-scale scenes, an accurate reference image is difficult to generate via investigation. Therefore, the reference images are not provided for the three scenes. However, the changed region can be roughly estimated by comparison.

Then, we introduce evaluation criteria to analyze the results of our method of change detection. Evaluating the result is a process of comparing the changed and unchanged pixels between the result and the reference image. The basic criteria are TP (changed pixels that are truly detected as changed), FP (unchanged pixels that are wrongly detected as changed), TN (unchanged pixels that are truly detected as unchanged), and FN (changed pixels that are wrongly detected as unchanged). The overall error (OE) is defined as the sum of FN and FP [62]

$$\text{OE} = \text{FN} + \text{FP}. \tag{13}$$

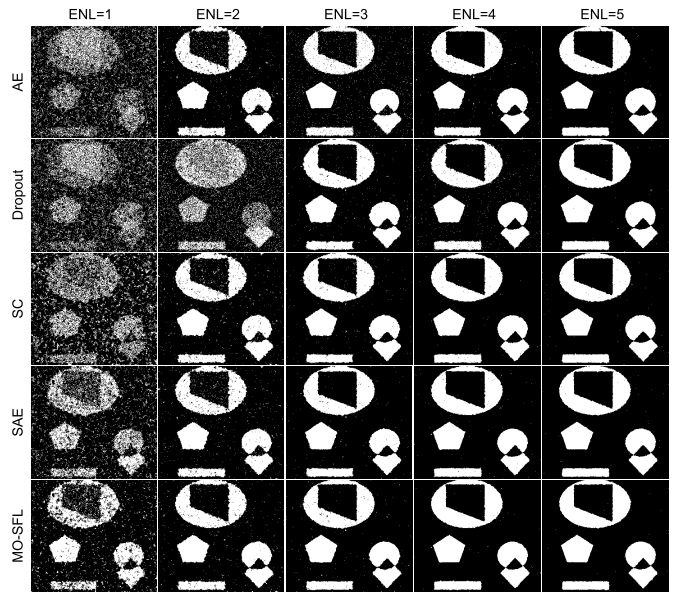


Fig. 12. Change detection results of different sparse learning methods on the simulate dataset.

The percentage correct classification (PCC) is defined as

$$\text{PCC} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}. \tag{14}$$

Another important metric is Kappa [63]. Kappa measures accuracy or agreement based on the difference between the error matrix and chance agreement. It is given by

$$\text{Kappa} = \frac{\text{PCC} - \text{PRE}}{1 - \text{PRE}} \tag{15}$$

where

$$\text{PRE} = \frac{(\text{TP} + \text{FP})(\text{TP} + \text{FN}) + (\text{FN} + \text{TN})(\text{TN} + \text{FP})}{(\text{TP} + \text{FP} + \text{FN} + \text{TN})^2}. \tag{16}$$

A result that is more close to the ground truth will result in lower OE, and higher PCC and Kappa.

There are three types of hyperparameters in SFCNet, including those in MOEA/D, network architecture parameters, and λ_A. We follow the parameter settings in [41] for MOEA/D. For network architectures, we set three hidden layers for all the experiments with 50, 30, and 20 hidden units of them, respectively. We set $\lambda_A = 0.1$ for all the experiments. To complete learning, we set six iterations for sparse feature learning and 100 epochs for fine-tuning.

B. Experiment on the Simulate Dataset

In this part, based on the simulate dataset with different noise levels, we compare the proposed SFCNet with AE and two popular sparse feature learning models, i.e., sparse coding (SC) and sparse AE (SAE) to demonstrate the superiority of MO-SFL adopted in SFCNet. AE is compared to demonstrate the effectiveness of multiobjective learning since one of the objectives in MO-SFL is that of AE. SC and SAE are both unsupervised learning models for extracting hierarchical sparse features. SC [64] aims to find a set of base functions among which the input data can be well

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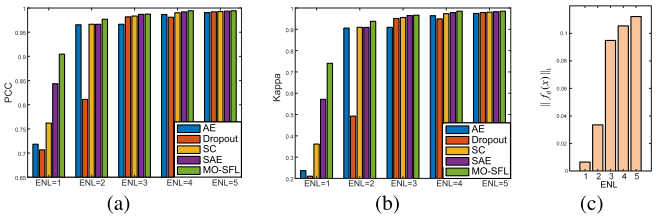


Fig. 13. Bar plots of evaluation results on the simulate dataset. (a) PCC. (b) Kappa. (c) l_1-norm of the learned feature.

reconstructed with as less base functions as possible. SAE [34] is based on the AE, which is a neural network model. The basic principle is also to generate efficient features that can represent the input data while removing redundant information with sparse codes. The two methods can both learn useful sparse features with an adjusted control parameter, which may be different with different input data. As introduced above, SFCNet with MO-SFL learning is adaptive, which can generate a satisfactory degree of sparsity without determining the adjusted control parameter. SFCNet with MO-SFL can well deal with the heterogeneous noise in SAR images. Moreover, as the widely used learning trick, Dropout [65] can also be taken as a sparse strategy. Therefore, the SFCNet with Dropout is also compared in this experiment.

The results on images of different noise levels are shown in Fig. 12. In general, better results can be achieved with higher ENL of the data. When ENL $= 1$, the images are severely corrupted by speckle noise. AE and Dropout fail to detect the correct changed region. Dropout was proposed to mitigate the overfitting of learning machines. The neurons are randomly dropped. Even though the output feature is also sparse, the sparseness is random. Other sparse learning methods, SC, SAE, and MO-SFL, are able to recognize the changed region. However, among them, SC and SAE are significantly influenced by the noise. The quantitative evaluation of these results with PCC and Kappa is shown as bar plots in Fig. 13(a) and (b). With less noise, the sparse learning methods achieve equivalent results. While when the images are severely corrupted by speckle noise, the results differ a lot where SFCNet with MO-SFL significantly outperforms the compared methods. In addition, to show the sparsity adaption of the MO-SFL, we plot the l_1-norm of the learned feature for each noise level in Fig. 13(c). When the images are severely corrupted by noise, the sparsity of the learned code is large to remove more redundant information, while, when there is less noise, the sparsity is lower to maintain more useful information.

C. Experiments on Small Patches of Real SAR Datasets

With the accurate reference images, we evaluate the proposed method adequately in this part. First, to highlight the adaptive sparse learning of SFCNet with MO-SFL, we compare SFCNet with MO-SFL and SAE (different sparsity is controlled by an important parameter λ). Then, evaluate the influence of the number of hidden layers in SFCNet, and highlight the effect of the proposed objective function. Finally, SFCNet is compared with some state-of-the-art methods.

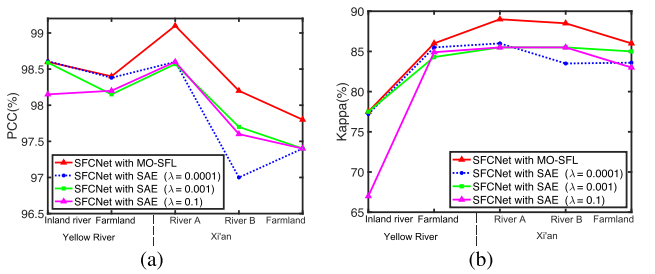


Fig. 14. Results of SFCNet with MO-SFL and SAE with different sparseness on the five small patches. The red triangle represents the result of SFCNet with MO-SFL, and the others represent SAE with different control parameters. (a) PCC. (b) Kappa.

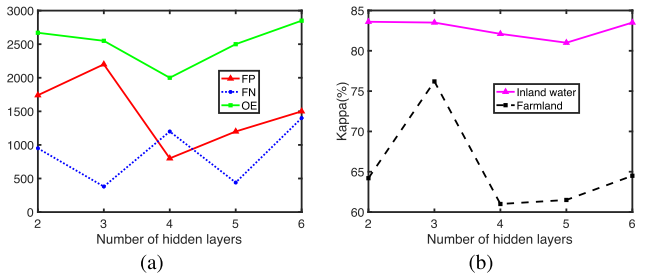


Fig. 15. Results of different numbers of hidden layers on the Yellow River dataset. (a) Relationship between FP, FN, OE, and the number of hidden layers on the inland water scene. (b) Relationship between Kappa and the number of hidden units on the scenes of inland water and farmland.

1) Comparison With SAE of Different Sparsity: Here, we set different control parameters λ of SAE and compare them with SFCNet with MO-SFL. The parameter is set to be 0.1, 0.001, and 0.0001, respectively, and the results on the five scenes with the criteria of PCC and Kappa are illustrated in Fig. 14. The horizontal axis denotes the five scenes, and the vertical axis denotes the evaluation criterion. It is obvious that, with different control parameters, the result differs a lot in some scenes, and the SAE is not stable with different images. The proposed method achieves better results in the five scenes, which demonstrates that the network SFCNet is able to adapt to different noise levels and achieves reasonable sparsity according to the input data.

2) Experiment on Number of Hidden Layers in SFCNet: With a DNN, the number of hidden layers may influence the performance of the network. Therefore, we set different numbers of hidden layers and evaluate the performance of the network. The number of hidden layers is set to 2–6, respectively, and obtain the FP, FN, and OE values on the scene of the inland river of the Yellow River dataset, as shown in Fig. 15(a). With fewer hidden layers, the network parameters may not well represent the input images and introduce more redundant information. With more hidden layers, some useful information may be lost. Therefore, when the number of hidden layers is 4, the network achieves the best performance for the scene of the inland river. In Fig. 15(b), the network performance is evaluated by the Kappa criterion on the scenes of the inland river and farmland of the Yellow River dataset. It can be found that, with different scenes, the optimal numbers of hidden layers are also different, e.g., 3, for the scene of farmland based on Kappa. However, the above experiments are implemented on the network with the same number of

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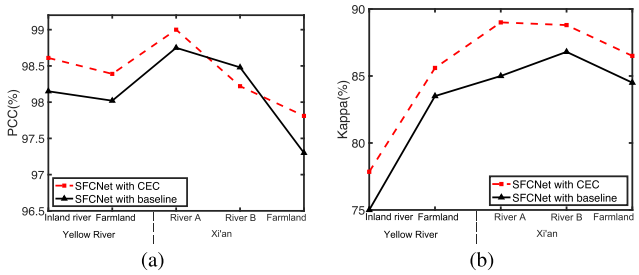


Fig. 16. Results of SFCNet with and without clustering regularization in the fine-tuning loss on five scenes of the Yellow River and Xi'an datasets. Baseline denotes the original objective function, and CEC denotes that with the clustering regularization. (a) PCC. (b) Kappa.

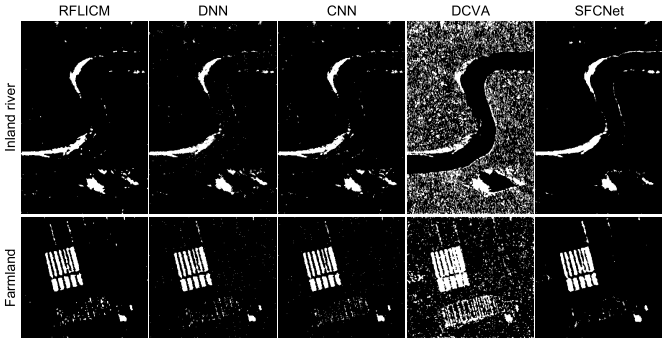


Fig. 17. Change detection results of different methods on the two small patches of the Yellow River dataset.

hidden units of 3, and the network achieves satisfactory results in different scenes.

3) Evaluation of the Proposed Objective Function: As introduced above, the clustering regularization term is introduced to the fine-tuning loss function to improve performance. By using the regularization term, the network can fully use all samples in addition to selected labeled ones. We then implement the network with and without the clustering regularization term on the five scenes and obtain the result evaluated by PCC and Kappa, as shown in Fig. 16, where baseline denotes the original objective function L_A in (9) and CEC denotes $\overline{L}_A$. It can be found that the network with the clustering regularization term performs better on most scenes from the PCC criterion. From the Kappa criterion, SFCNet with the cluster regularization term can achieve a better performance than the baseline. This experiment demonstrates the superiority of the clustering-based fine-tuning loss proposed in SFCNet.

4) Comparison With State-of-the-Art Approaches: In this part, we compare the proposed method SFCNet with several change detection methods to verify the effectiveness of the proposed SFCNet on the five scenes. The first one is based on wavelet fusion and improved fuzzy local information c-means algorithm called RFLICM [26]. Two types of DIs were fused to generate a complemented DI, and RFLICM was proposed to generate the final change map. Then, a DNN-based change detection method [34] and a CNN-based one [20] are compared since the proposed method is also based on neural networks. Finally, we choose the most recent change detection method, the deep change vector analysis (DCVA), [66] based on deep features for comparison. The results on the Yellow River and Xi'an datasets are shown in Figs. 17 and 18, respectively.

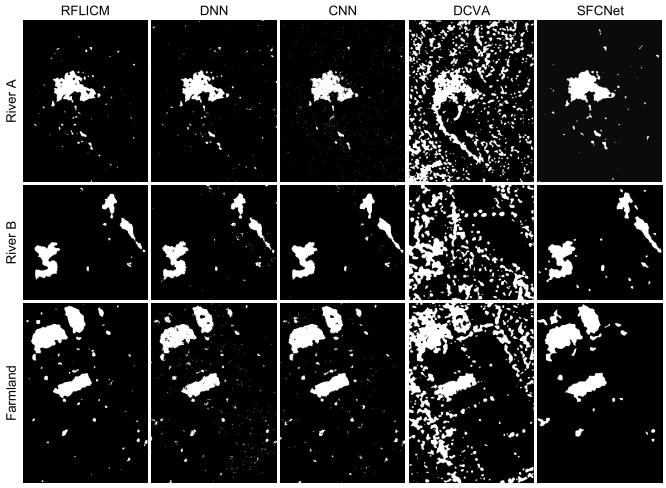


Fig. 18. Change detection results of different methods on the three small patches of the Xi'an dataset.

TABLE I

EVALUATION RESULTS ON THE TWO SCENES OF THE YELLOW RIVER DATASET

Scenes	Methods	FP	FN	OE	PCC(%)	Kappa
Inland river	RFLICM	941	804	1745	98.04	0.8219
	DNN	989	679	1668	98.13	0.8270
	CNN	976	669	1645	98.15	0.8295
	DCVA	99	8590	9049	89.84	0.4893
	SFCNet	720	704	**1424**	**98.40**	**0.8562**
Farmland	RFLICM	862	1300	2162	98.33	0.7497
	DNN	821	1411	2232	98.27	0.7458
	CNN	824	1317	2141	98.34	0.7536
	DCVA	968	29842	30810	76.15	0.1248
	SFCNet	763	1124	**1887**	**98.54**	**0.7797**

Among the compared methods, DCVA achieves the worst results on all the scenes. Even though it was proposed based on features extracted from DNN, the model was not specially designed for denoising. The changed map obtained by DCVA is corrupted by many white spots, and many changed pixels are miss detected. This demonstrates the effect of sparse learning in SFCNet. RFLICM segments the DI by using the local spatial and spectral information. It considers the similarity between the central and neighbor pixels. Therefore, it is robust to noise to some extent. However, this may lead to the loss of image details, such as lines and contours. Then, the margin of the changed regions cannot be well retained by RFLICM in the changed map. Then, the DNN-based method deals with the noise by extracting useful features via RBM and selecting samples by a neighborhood-based selection criterion. Stacked RBM is able to extract abstract information, which removes the redundant noise. The CNN-based method was designed for multiple change detection, and it can also accurately detect the changes from SAR images. The quantitative evaluations are listed in Tables I and II. From these criteria, SFCNet improves the DNN-based method by incorporating MO-SFL and the clustering regularization on the fine-tuning loss. It adaptively learns features and reduces the influence of the selected samples. From the Xi'an dataset, the DNN-based method cannot outperform the RFLICM. However, with the sparse feature and the clustering regularization, the performance can

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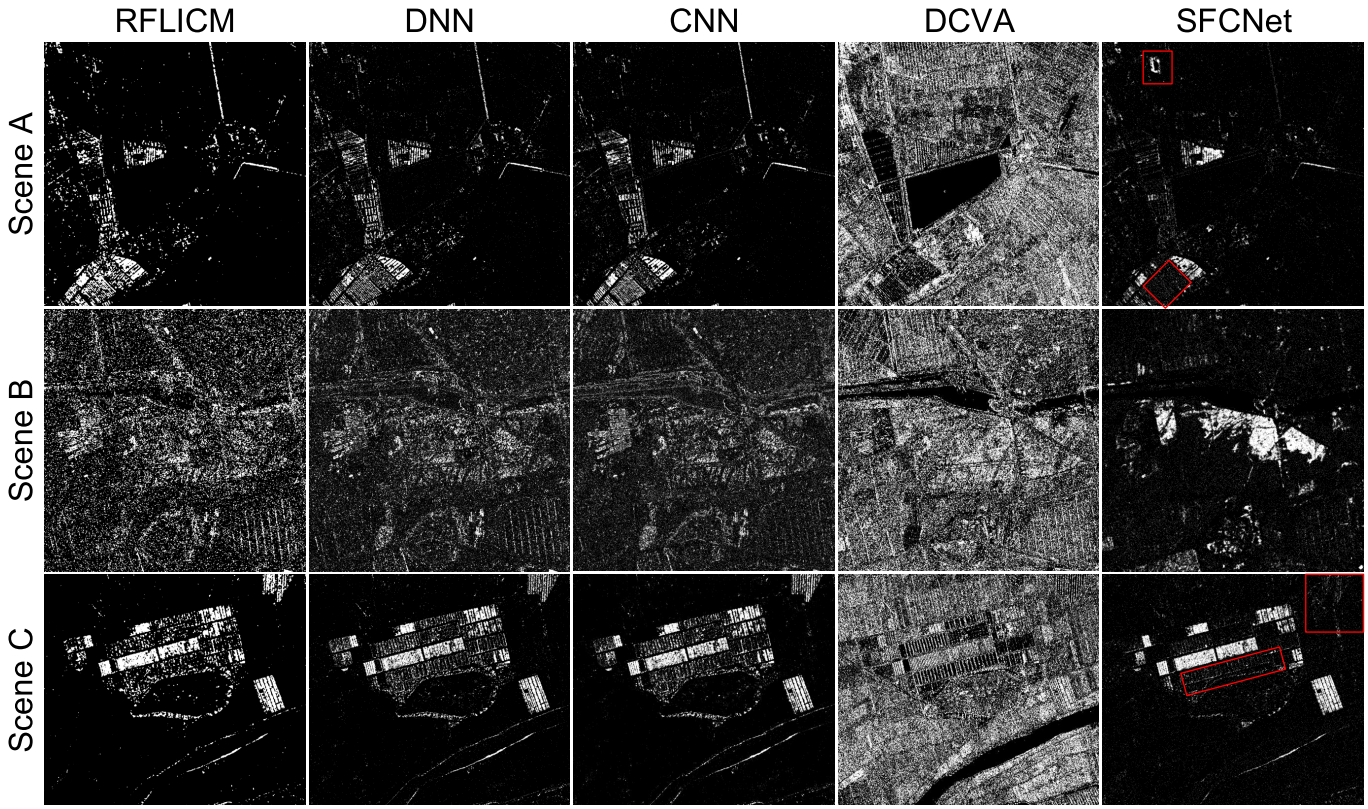


Fig. 19. Change detection results of different methods on the three large patches of the Yellow River dataset.

TABLE II
EVALUATION RESULTS ON THE THREE SCENES OF THE XI'AN DATASET

Scenes	Methods	FP	FN	OE	PCC(%)	Kappa
River A	RFLICM	974	731	1706	98.55	0.8451
	DNN	1417	509	1926	98.36	0.8150
	CNN	1349	521	1870	98.41	0.8216
	DCVA	1517	21764	23281	80.22	0.2092
	SFCNet	726	462	**1188**	**98.99**	**0.8918**
River B	RFLICM	836	271	1107	97.62	0.8509
	DNN	1401	155	1556	96.66	0.7736
	CNN	1427	86	1513	96.75	0.7773
	DCVA	2341	8126	10467	77.52	0.1686
	SFCNet	714	104	**818**	**98.24**	**0.8893**
Farm-land	RFLICM	1380	532	1912	97.40	0.8258
	DNN	1506	842	2348	96.81	0.7890
	CNN	1354	794	2148	97.08	0.8085
	DCVA	1216	10683	11899	83.85	0.3899
	SFCNet	740	876	**1616**	**97.81**	**0.8629**

be significantly improved by SFCNet. A reasonable balance between the false and miss alarms can be obtained by the proposed method.

D. Experiments on Large-Scale Scenes of the Yellow River Dataset

To further evaluate the proposed method, three large-scale patches selected from the Yellow River dataset are used. However, with a large-scale covered region, it is difficult to mark the accurately changed region. However, some changed regions can be recognized by artificial comparison. The results of the three scenes are shown in Fig. 19. Due to the severe influence of speckle noise, some compared methods miss the changed region while highlighting unchanged regions, such as the regions surrounded by the red box. The superiority of SFCNet can be demonstrated from the results of scene B where the compared methods cannot well recognize the changed region, while the noise is significantly highlighted. Both SFCNet and DCVA use learned features for change detection. DCVA is pretrained by large-scale labeled images to fully represent the images, while, with severely corrupted noise, it fails to correctly detect the changed region. SFCNet is pretrained by the MO-SFL, which is designed to represent input data with sparse code. This demonstrates the denoising capability of the proposed SFCNet.

V. CONCLUSION AND FUTURE WORK

This article proposes an SFCNet for change detection in SAR images. The method is based on the framework that directly processes the multitemporal SAR images and obtains the change detection result without generating the DI. Therefore, the framework has the superiority of avoiding the influence of the quality of the DI on the changed map. However, the main difficulty in dealing with SAR images is to reduce the impact of speckle noise. Specifically, with different SAR images, the noise level is also different. Therefore, we pretrain SFCNet by an MO-SFL to adaptively learn the sparse features. Then, it can adapt to data with different noise levels. Since the network needs the labeled samples to learn the semantics of changed and unchanged regions, some samples are selected from the result obtained by other methods. However, with many wrongly labeled samples, the selection criteria influence the final change detection result a

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lot. Therefore, we introduce a clustering-based regularization term into the objective function of fine-tuning in order to learn discriminative and compact features, and reduce the impact of selection criteria. Experiments on the simulate and real SAR datasets demonstrate the effectiveness of the proposed method. It achieves stable results on different datasets.

In the future work, we will focus on more difficult change detection problems, such as detecting changes from multisensor images and multiview images, which are obtained with different view angles and propose efficient models for such problems.

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